# VerusCite V1 Benchmark: Citation Verification Accuracy

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# 1 The Problem

Fabricated citations in academic papers have grown sharply since the widespread adoption of generative AI writing tools. A Lancet audit of 2.5 million biomedical papers found hallucinated reference rates rising 12-fold between 2023 and early 2026 (Topaz et al. 2026). Separate estimates project roughly 147,000 hallucinated citations across arXiv, bioRxiv, SSRN, and PMC in 2025 alone (Zhao et al. 2026).

Existing peer review and editorial safeguards largely fail to catch these errors before publication. The problem spans disciplines: biomedical research, AI/ML conferences, legal filings, and government reports.

While AI use is driving a surge in publications (see A. Wheeler 2026 for a practical overview), the same large language models that create this problem can also be used to verify citations at scale. VerusCite (veruscite.com) is a tool to automatically extract and verify citations from uploaded documents. This report documents the accuracy of the V1 verification pipeline against a hand-labeled ground truth corpus.

This tool is meant to be a quick and cheap approach to scan a bibliography for a paper, and determine if a citation is correct, has minor errors, or appears to have more serious issues, such as an AI generated hallucination. This paper serves as reference to the overall architecture and design of the project, as well as a source of false positive and negative rates on a benchmarked corpus of real papers.

# 2 Approach

## 2.1 Design Philosophy

Even in documents with hallucinated citations, the majority of references are legitimate. A tool with a high false positive rate – flagging real papers as hallucinations – would waste a reviewer's time chasing false alarms rather than finding actual problems. VerusCite prioritizes **precision over recall**: it is better to miss some hallucinations than to incorrectly flag verified citations.

The target false positive rate for hallucination flags is below 1%. Recall for detecting actual hallucinations is around 70-80%. This trade-off means a human reviewer spends most of their time on citations that genuinely need attention rather than dismissing false reports.

The application is designed for human-in-the-loop review. Each citation's verification status can be manually overridden, and the interface surfaces the reasoning behind each determination so reviewers can make informed decisions quickly.

Similar to work in Pangram, I believe that it is necessary to have the tool have as few of false positives as possible (Emi 2024). While this report will show that those false positives are not zero (any realistic tool should report a certain error rate), the design still prioritizes keeping them as low as possible.

## 3 Pipeline Architecture

VerusCite processes documents in two stages:

1. **Citation Extraction** – An LLM reads the document text and identifies individual references, parsing out structured fields (title, authors, year, journal, DOI).
2. **Citation Verification** – Each extracted citation is checked against external sources to determine whether it exists as described.

For citation extraction, the tool uses text searching across the document (if a PDF or word document is uploaded) to attempt to determine the location of the reference section. This tool currently works with reference sections combined in an article, it does not work with footnote style citations (as is common in law articles).

If uploading a bibliography file, such as a Bibtext `.bib` file, a `.ris` file, or an Endnote `.enw` file, the tool uses text tools to automatically extract out the reference information.

Once the reference section is identified, the reference section is then chunked into smaller portions, and uses a served LLM model (see later benchmark results) to extract out the citations into a structured sections. This means the LLM is only sent the references for the document, it is not sent other parts of the manuscript.

Each extracted citation is then checked independently. First the citation is checked against the Crossref database. If it is not found in Crossref, then an agentic tool uses web-search and fetch to verify whether the citation exists in external sources.

This independence means that in a document with some errors, other documents are not falsely assumed to be hallucinated. It also allows processing the documents in parallel, to return the results back faster.

## 3.1 Models Considered

Because the task involves web-search, when checking citations that are not verified via Crossref, this limits the potential pool of LLM providers. Here I consider two major companies:

- Gemini family models
- OpenAI family models

In particular, I focus on the cheaper models (flash-lite and flash for Gemini, Nano/Luna for OpenAI) with minimal reasoning. I additionally consider these same models, but using the web search tools provided by *Perplexity*.

I do not consider Anthropic, as even their cheapest model (Haiku) is considerably more expensive than the models I evaluated here. I only partially evaluated AWS Nova-2 lite with their web search (latency was too long). Grok 4.3 did well in the benchmark tests but was considerably more expensive.

Perplexity also hosts open-weight checkers. **DeepSeek V4 Flash** (`perplexity/deepseek-v4-flash-0731`) is **not included** in the tables: repeated full-corpus attempts, including a 2026-08-18 retry, consistently returned HTTP **500 stream / internal_error** responses (and 180-second timeouts) from Perplexity's Agent API, so there is no usable baseline. **Nemotron 3.5 Lightning** (`perplexity/nemotron-3.5-lightning-30b-a3b`) did complete a full run and is included as a low-cost option, but latency is very long (see Cost below). The 2026-08-18 table also adds Perplexity-hosted **Gemini 3.5 Flash Lite** and **Gemini 3.7 Flash**.

The 2026-09-24 Perplexity runs use a different tool setup from the earlier Perplexity rows. Each model is offered `fetch_url` (up to 5 URLs) in addition to `web_search`, and is allowed up to 5 research steps, so it can search, open a candidate publisher page, and search again. Earlier Perplexity runs had search only and a single research step. These runs also compare Perplexity's two search tiers: standard search ($2.50 per 1,000 calls) and **fast** search ($1 per 1,000 calls). Rows using fast search are labeled `perplexity (fast)` in the tables.

Extraction does not need web search, and so while I evaluated many other models on Bedrock (such as the open source GPT and Gemini models), they were not as accurate as the frontier served models. Extraction takes around 2 to 3 cents per document, so at this time I am not concerned about cost, although in the future likely even smaller open source models (especially if fine tuned) will be sufficient for the extraction task.

## 3.2 Verification Categories

The checker assigns each citation one of four statuses:

| Status | Meaning |
|---|---|
| **Verified** | Citation confirmed to exist via CrossRef or web search |

| Status | Meaning |
|---|---|
| **Minor Error** | Citation likely exists but has metadata discrepancies (wrong year, volume, pages) |
| **Hallucination** | No evidence the cited work exists as described |
| **Not Found** | Unable to confirm or deny – includes bare URL citations and paywalled/very recent works with insufficient search results |

Many citations have minor errors (perhaps on the magnitude of 5% in my personal assessment). These include many innocuous things that are likely human introduced errors, like a year off or page numbers wrong.

The major different between a hallucination and a minor error is that hallucinations tend to have whole cloth differences that are difficult to explain. Such as journals entirely replaced. For example, Canessa et al. (2026) has the citation:

> Elrod, Linda D. (2006). "A Move in the Right Direction? Best Interests of the Child Emerging as the Standard for Relocation Cases". Journal of the American Academy of Matrimonial Lawyers 15, pp. 1-54.

This is a real paper, but should be cited as:

> Elrod, Linda D. (2006). "A Move in the Right Direction? Best Interests of the Child Emerging as the Standard for Relocation Cases". Journal of Child Custody 3, 29–61. https://doi.org/10.1300/J190v03n03_03

While the Journal of the American Academy of Matrimonial Lawyers does exist, there is no combination of volumes and page numbers that could reasonably be confused with the article in the Journal of Child Custody.

While no automated tool can never find 100% proof that a citation was hallucinated via a generative AI tool, it is difficult to construct a scenario where a bibliography had such errors manually generated, especially the volume and page numbers that do not exist in any comparable document for the Elrod citation.

Hallucinations can also include when multiple authors are incorrect, but author name errors are much more common, so they are less likely to be flagged. For an example of a minor error Mekonen (2026) has the reference:

> 33. Croft TMA, Allen CK, Arnold F, Assaf S, Balian S. Guide to DHS Statistics: DHS-7 (version 2). Rockville, MD: ICF. 2020.

The actual author list includes an Aileen M.J. Marshall, whose initials were concatenated into the first author (see page 2 at https://dhsprogram.com/pubs/pdf/DHSG1/Guide_to_DHS_Statistics_DHS-7_v2.pdf).

Minor errors are incredibly prevalent. Any tool intended to do the same task needs to effectively distinguish between minor errors to be able to identify actual hallucinated citations.

# 4 Benchmark

The V1 validation corpus contains **2288** hand-labeled citations across **36 documents**. Sources are intentionally heterogeneous:

- Papers with known hallucinations identified by others on social media (Chris Carothers on X, David Buil-Gil on BlueSky)
- GPTZero's NeurIPS 2025 audit (GPTZero 2026)
- Papers flagged by Reviewer3 (Reviewer3, n.d.)
- A ChatGPT deep-research output where all 12 citations are fabricated (Jacques et al. 2026)
- Clean papers (my own dissertation excerpts, open-access criminology, PLoS ONE articles)
- MDPI and preprint samples

The corpus spans categories in social sciences, physics, mathematics, as well as many different formats, including many different pre-print and journal format examples.

Of the 2288 citations, 1889 (82.6%) are verified, 204 (8.9%) have minor errors, 141 (6.2%) are hallucinations, and 54 (2.4%) are not found. The corpus skews heavily toward verified citations (as real documents do), which makes false positive rate the critical metric.

There are two different metrics a user should be interested in. First is whether the tool can actually extract out the references accurately. This is non-trivial in and of itself – a skills based approach submitting a large document with an extensive bibliography is likely the generate more errors in this steps.

The second metric is in terms of false positives (e.g. a good citation is flagged as a hallucination), and recall rates (of all the hallucinations, how many does the tool capture). Below are those metrics for this corpus on the current VerusCite tool across different LLM providers.

## 4.1 Extraction Results

Citation extraction uses an LLM to parse the document text and identify individual references. The table below shows extraction accuracy against the ground truth. There are two types of errors that can occur, you can miss a reference, or the tool can add in a reference.

Table 1: Extraction accuracy by model

| Model | Real | Correct | Missing | Extra | Rate (%) | Wall min | Docs/min | Cost |
|---|---|---|---|---|---|---|---|---|
| gemini-3.1-flash-lite | 2288 | 2285 | 3 | 2 | 99.9 | 4.9 | 7.4 | 1.15 |

Table 1: Extraction accuracy by model

| Model | Real | Correct | Missing | Extra | Rate (%) | Wall min | Docs/min | Cost |
|---|---|---|---|---|---|---|---|---|
| gemini-3.1-flash-lite | 2288 | 2275 | 13 | 4 | 99.4 | 5.6 | 6.4 | 1.11 |
| gemini-3.1-flash-lite | 2288 | 2272 | 16 | 1 | 99.3 | 4.4 | 8.1 | 1.11 |
| gemini-3.5-flash-lite | 2288 | 2286 | 2 | 2 | 99.9 | 4.0 | 8.9 | 2.15 |
| gpt-5.4-nano | 2288 | 2285 | 3 | 3 | 99.9 | 12.2 | 3.0 | 0.88 |
| gpt-5.6-luna | 2288 | 2284 | 4 | 5 | 99.8 | 8.2 | 4.4 | 0.84 |
| gpt-5.6-luna | 2288 | 2283 | 5 | 4 | 99.8 | 9.0 | 4.0 | 0.74 |
| gpt-5.6-luna | 2288 | 2282 | 6 | 7 | 99.7 | 7.3 | 4.9 | 0.74 |

Extraction is very accurate across models. Gemini 3.1 Flash Lite (using pypdfium2 to read in the text, we additionally considered liteparse with markdown) correctly extracts over 99.8% of citations at under $0.03 per paper for the full corpus. The few "extra" rows are typically page footers, footnotes, or duplicate listings that the LLM picks up in addition to the actual bibliography.

I also tested `liteparse` with the markdown export format to extract out the text from PDFs, which tended to produce slightly more artifacts – it does not distinguish line breaks for individual citations as cleanly as pypdfium2, producing a few more missed or merged entries. Results were largely similar overall (and I may migrate to using liteparse with markdown in the future).

Both the extraction part of the pipeline and the citation check part of the pipeline have fallback models, as it is common for these LLM providers to have downtime with models. The current production version of VerusCite uses OpenAI's gpt-5.6-luna as the primary extraction model, with Google's Gemini 3.5 Flash Lite as the fallback. In the 2026-08-27 full-corpus runs, Luna correctly extracted 2,283 citations with 5 missing and 4 extra, while Gemini correctly extracted 2,275 with 13 missing and 4 extra. Luna also cost less for the corpus, although its run took longer.

Generally the prompts are constructed in a way that advanced reasoning is not necessary. So going up to larger reasoning models (and expanding the reasoning budget) does not result in higher accuracy. For OpenAI, you can see that the recent luna model did not result in any more accurate results than nano.

## 5 Citation Checking Results

The citation checking results break down the benchmark against both the model, and the provider. When Perplexity is the provider, it means the tool called Perplexity servers (using their internal web search), whereas if the provider is gemini it uses Googles web search tool, or OpenAI uses its internal web search tool. Otherwise all prompts are the same across each of the models.

The citation checking results input in correct data – extraction and checking are done in independent benchmarks. So each of the 2288 citations, it is independently run though the checking benchmark, swapping out the model used.

Where the same model/provider appears more than once, those are **independent full-corpus re-runs** (different dates), not duplicates of one output. Repeat runs are intentional: agentic web search and LLM classification are stochastic, so multiple runs show run-to-run variance in false positives, recall, cost, and wall time. Accuracy tables include a **Date** column so repeats are distinguishable.

Table 2: Citation Checking accuracy by model/provider (repeat rows are independent re-runs)

| Model | Provider | Date | FP Hall. | FP Minor | Hall. Recall | NV Recall |
|---|---|---|---|---|---|---|
| gemini-3.5-flash-lite | gemini | 2026-07-29 | 0.1% (2) | 0.8% (15) | 63.1% (89/141) | 71.9% (286/398) |
| google/gemini-3.1-flash-lite | perplexity | 2026-07-30 | 0.4% (7) | 0.7% (13) | 69.5% (98/141) | 82.7% (329/398) |
| google/gemini-3.1-flash-lite | perplexity | 2026-08-03 | 0.4% (7) | 1.2% (22) | 64.5% (91/141) | 80.2% (319/398) |
| google/gemini-3.1-flash-lite | perplexity | 2026-08-13 | 0.4% (7) | 1.1% (20) | 47.5% (67/141) | 80.2% (319/398) |
| google/gemini-3.1-flash-lite | perplexity | 2026-08-13 | 0.6% (11) | 0.9% (17) | 68.8% (97/141) | 79.1% (315/398) |
| google/gemini-3.5-flash-lite | perplexity | 2026-08-18 | 0.1% (1) | 0.5% (9) | 46.1% (65/141) | 67.1% (267/398) |
| google/gemini-3.5-flash-lite | perplexity (fast) | 2026-09-24 | 0.3% (5) | 0.4% (7) | 59.6% (84/141) | 70.4% (280/398) |
| google/gemini-3.5-flash-lite | perplexity | 2026-09-24 | 0.2% (4) | 0.5% (10) | 60.3% (85/141) | 71.6% (285/398) |
| google/gemini-3.7-flash | perplexity | 2026-08-18 | 0.2% (4) | 0.5% (9) | 72.3% (102/141) | 78.9% (314/398) |
| google/gemini-3.7-flash | perplexity | 2026-08-21 | 0.1% (2) | 0.6% (12) | 68.1% (96/141) | 81.2% (323/398) |
| gpt-5.4-nano | openai | 2026-07-30 | 0.2% (3) | 2.2% (42) | 60.3% (85/141) | 88.4% (352/398) |
| gpt-5.6-luna | openai | 2026-07-30 | 0.2% (3) | 2.5% (47) | 74.5% (105/141) | 88.7% (353/398) |
| gpt-5.6-luna | openai | 2026-08-03 | 0.1% (2) | 2.8% (52) | 73.8% (104/141) | 89.2% (355/398) |
| gpt-5.6-luna | openai | 2026-08-13 | 0.2% (4) | 2.4% (45) | 72.3% (102/141) | 88.7% (353/398) |
| gpt-6-luna | openai | 2026-09-22 | 0.4% (7) | 2.3% (43) | 63.8% (90/141) | 88.9% (354/398) |
| openai/gpt-5.4-nano | perplexity | 2026-07-30 | 0.4% (7) | 2.5% (47) | 33.3% (47/141) | 84.9% (338/398) |
| openai/gpt-5.6-luna | perplexity | 2026-08-01 | 0.2% (4) | 2.2% (41) | 44.7% (63/141) | 87.7% (349/398) |
| openai/gpt-5.6-luna | perplexity | 2026-08-13 | 0.2% (3) | 2.0% (38) | 44.0% (62/141) | 87.7% (349/398) |

Table 2: Citation Checking accuracy by model/provider (repeat rows are independent re-runs)

| Model | Provider | Date | FP Hall. | FP Minor | Hall. Recall | NV Recall |
|---|---|---|---|---|---|---|
| openai/gpt-6-luna | perplexity (fast) | 2026-09-24 | 0.4% (7) | 3.2% (61) | 70.9% (100/141) | 90.2% (359/398) |
| openai/gpt-6-luna | perplexity | 2026-09-24 | 0.4% (8) | 2.4% (45) | 63.8% (90/141) | 87.4% (348/398) |
| openai/gpt-6-luna | perplexity (fast) | 2026-11-12 | 0.3% (5) | 2.9% (55) | 70.9% (100/141) | 89.7% (357/398) |
| perplexity/nemotron-3.5-lightning-30b-a3b | perplexity | 2026-08-13 | 0.1% (1) | 0.5% (10) | 30.5% (43/141) | 63.6% (253/398) |

## 5.1 False Positives

For key metrics, there are a total of 1889 correct citations in the corpus. So false positives (FP) rates are of those 1889 correct citations, how many were falsely flagged by the tool.

The current FP rates for hallucinations are well under 1%, and vary between 1 to 11 total false positives across the different models in this corpus (and can shift by a few counts between re-runs of the same model). Hallucination false positives are often attributable to idiosyncratic web search results. (The agentic tools often will return a response, even if the web search tool is currently inaccessible.) Verified false-positive rates on Perplexity are broadly similar to the same model on the provider's own web-search stack, but **hallucination recall is not** in the earlier search-only setup: Perplexity-hosted OpenAI models (`openai/gpt-5.4-nano`, `openai/gpt-5.6-luna`) under-detect true hallucinations relative to OpenAI direct. Adding URL fetch and multiple research steps closes that gap (see Recall below).

For an example of a false positive hallucination in this particular run, my dissertation has the citation:

> Pearl, J. (2000). Causality: models, reasoning and inference, Volume 29. Cambridge Univ Press.

And 5.4 nano lists as reasoning for classifying this as a hallucination:

> Upstream verdict: book exists and matches author/title/year/publisher, but the cited "Volume 29" element is not supported by the primary publisher/library records surfaced in the tool results and is treated as fabricated metadata; therefore status mapped to hallucination.

This behavior is not consistent, either within 5.4. nano or across other models. (This most often returns a "minor error", although my ground truth I have this as "verified". Some sources list the published year as 2001, causing additional issues beyond just the "Volume 29" addition.) I have intentionally avoided K-shot examples in the prompt, so there is as little

leakage as possible and the prompts should better generalize to out of sample data. But given the stochastic nature of LLMs, some errors will ultimately occur.

FP rates for minor errors are more prevalent, being close to 1% for the Gemini models, but over 2% for the OpenAI models. Production defaults are summarized in Current Production Configuration: primary checker is **Perplexity-hosted `openai/gpt-6-luna`** with fast search, with **OpenAI direct `gpt-6-luna`** as the fallback. The production primary has a higher minor-error FP rate (3.2%) than the Gemini configurations.

The general approach I took was to evaluate when both OpenAI and Google models returned false positives. While these do happen in the corpus, in some cases they are inevitable, as CrossRef or other online data provides conflicting information. For one example citation:

> [CS22] T. Cieśla and M. Sabok. Measurable Hall's theorem for actions of abelian groups, J. Eur. Math. Soc., 24 (2022), 2751-2773 (cit. on pp. 3, 44)

Consistently produces a "minor error" category in the tool due to the citation year. Going to the website, the citation year is correctly 2022. But crossref lists the publication year as 2021.

Many of the false positives are minor errors like this (and likely some of the labels in the ground truth should be updated – feel free to contact me if you believe an articles classification is not correct in the ground truth). False positives for not found tend to be due to unreliable web search, and so are intermittent and not consistent due to the tool.

## 5.2 Recall

Recall is the proportion of the true errors that are captured by the current tool. These are more variable across the different tools, with OpenAI models having greater recall (which comes with more false positives). Hallucination recall for the Gemini configurations is typically in the **60–70%** range, but one Perplexity 3.1 flash-lite re-run (2026-08-13) dropped to **47.5%** (67/141) while a later same-day re-run recovered to **68.8%** (97/141). Across three independent full-corpus runs, **OpenAI direct `gpt-5.6-luna`** lands at about **72–75%** (102–105/141), while earlier production **Perplexity `google/gemini-3.1-flash-lite`** is usually about **65–70%** (91–98/141) aside from that one low re-run. The two Perplexity **`google/gemini-3.7-flash`** runs bracket that range: the 2026-08-18 run had **72.3%** hallucination recall (102/141) and **78.9%** not-verified recall (314/398), while the 2026-08-21 run had **68.1%** hallucination recall (96/141) and **81.2%** not-verified recall (323/398). Both kept false positives low. Perplexity **`google/gemini-3.5-flash-lite`** on 2026-08-18 also kept false positives low (0.1% hallucination FP, 0.5% minor FP) but hallucination recall dropped to **46.1%** (65/141), so it is not the production model. The same luna weights on **Perplexity** (`openai/gpt-5.6-luna`) stay near **44–45%** hallucination recall (62–63/141) across the 2026-08-01 and 2026-08-13 runs, with not-verified recall still high (about 88%) because many misses land in minor error or not found rather than verified. Perplexity `openai/gpt-5.4-nano` shows the same pattern

even more sharply (33% hallucination recall). A Perplexity **Nemotron 3.5 Lightning** run is much cheaper but weaker on errors (**30.5%** hallucination recall, **36.9%** minor-error recall) and is slow: about **12 minutes mean per paper** and **89 minutes** full-corpus wall. Provider web-search quality matters as much as the base model for this task.

The search-only, single-step setup appears to explain much of that Perplexity OpenAI gap. With `fetch_url` and up to 5 research steps on 2026-09-24, Perplexity **`openai/gpt-6-luna`** reached **70.9%** hallucination recall (100/141) with fast search and **63.8%** (90/141) with standard search. Not-verified recall was 87–90% in both runs. The same model through OpenAI direct (2026-09-22) had **63.8%** hallucination recall (90/141), down from 72–75% for direct `gpt-5.6-luna`. Under the same Perplexity setup, `google/gemini-3.5-flash-lite` improved to about **60%** hallucination recall (84–85/141, up from 46.1% on 2026-08-18), but not-verified recall stayed near 70–72%.

The fast search tier did not reduce accuracy in either pair. For luna, the fast run had higher hallucination recall (70.9% vs 63.8%) and a higher minor-error FP rate (3.2% vs 2.4%). For 3.5 flash-lite, the two tiers were within a few citations of each other on every metric. Each comparison is a single pair of runs, and differences of this size also occur between same-configuration re-runs, so the results support equivalence rather than an advantage for fast search.

That re-run spread is the main reason tables keep multiple rows for the same model: FP minor for Perplexity 3.1 flash-lite moved from 0.7 (13) to 1.2% (22) between the July and early-August runs, then sat at 0.9–1.1% on the 2026-08-13 pair, while OpenAI luna hallucination FP stayed at 0.1–0.2% and hallucination recall stayed near 72–75%. Cost and wall time also drift with provider load and search volume (see the cost table).

Minor error recall is similarly lower for Gemini direct 3.5 flash-lite, at 52.7%. OpenAI direct has higher recall (often higher than 70%) for minor errors. Not found recall is near perfect across all models.

The final category, not verified, collapses the categories of minor error, hallucination, and not found. So if many hallucinations were classified into minor error, the direct recall rates would be low, but not verified (which will typically trigger a human review) would still be high. These are consistently near or over 80% across Gemini and OpenAI runs, with the exception of Gemini-direct 3.5 flash lite (mostly low minor-error recall), Perplexity 3.5 flash lite (67.1% not-verified recall), and Nemotron (63.6% not-verified recall). Perplexity 3.7 flash was **78.9%** on 2026-08-18 and **81.2%** on 2026-08-21.

## 5.3 Current Production Configuration

Production VerusCite uses the following defaults for citation checking:

| Role | Provider | Model |
|---|---|---|
| **Primary checker** | Perplexity (fast search + fetch, up to 5 steps) | `openai/gpt-6-luna` |
| **Fallback checker** | OpenAI (direct) | `gpt-6-luna` |

The current production checker is **GPT-6 Luna with Perplexity fast web search and URL fetch** (`openai/gpt-6-luna`). On the 2026-09-24 full-corpus run it had **70.9%** hallucination recall (100/141), the highest hallucination recall since the direct `gpt-5.6-luna` runs, and **90.2%** not-verified recall (359/398). It cost **$3.75** for the corpus (about **$0.10 per paper**), roughly a third of the previous primary (Perplexity `google/gemini-3.7-flash`, about $11.95 / $0.33 per paper after the search-price correction described below). The trade-off is a higher minor-error false-positive rate: **3.2%** (61) compared with 0.5–0.6% for 3.7 flash. Hallucination FP stayed low at **0.4%** (7). Full-corpus wall time is also longer, about 44 minutes compared with 24–30 minutes for 3.7 flash. A minor-error flag asks the user to compare metadata and does not claim fabrication, so I accept more minor-error flags in exchange for higher hallucination recall at lower cost.

The 2026-11-12 run re-checks the same configuration after the HALLMARK-driven changes listed in the Changelog. Hallucination recall was unchanged at **70.9%** (100/141), not-verified recall was **89.7%** (357/398), and minor-error recall was **74.4%** (151/203, compared with 148/203). False positives were slightly lower: **0.3%** (5) hallucination and **2.9%** (55) minor error. Cost was **$3.85** for the corpus and wall time was about 49 minutes. Differences of this size also occur between same-configuration re-runs, so I read this as no regression on the V1 corpus, not as an improvement.

Google's Gemini web-search stack caps searches at **1,500 per day across all tiers**, which is too low for multi-user production load (a single large paper can consume dozens of searches after Crossref misses). Perplexity does not impose that daily search cap, so agentic verification can keep running under concurrent documents. Gemini **direct** 3.5 flash-lite is still one of the fastest configurations in the table below (about 16 minutes full-corpus wall), but the search quota makes it unsuitable as the default production path. It is retained in the benchmark for comparison.

**Why OpenAI luna as backup.** Direct OpenAI `gpt-6-luna` is the fallback when Perplexity is down or returns persistent errors. On 2026-09-22 it had 0.4% hallucination FP (7), 2.3% minor-error FP (43), **63.8%** hallucination recall (90/141), and **88.9%** not-verified recall, at about **$12.49** ($0.35 per paper). That is lower hallucination recall than the earlier direct `gpt-5.6-luna` runs (72–75%), but it keeps the fallback on the same model family as the primary. Nemotron on Perplexity is cheap (about $4.27 full corpus after the search-price correction), but latency is very long (about 12 minutes mean per paper, 89 minutes full-corpus wall) and error recall is much weaker, so it is not used.

Extraction production uses OpenAI `gpt-5.6-luna` as the primary model, with Google Gemini 3.5 Flash Lite as the fallback.

## 5.4 Cost Breakdown for Citation Checking

For the 2288 total citations, approximately 1310 citations were verified via Crossref after extraction (about 57%) for each of the models. These are largely automated, and so incur no additional LLM cost. When those fail however, an agent based LLM tool needs to use web search and fetch to identify whether the citation is correct. Thus costs incur for both token usage as well as web search. Web search costs $2.50 per 1000 searches on Perplexity's Agent API (or $1 per 1000 with fast search, plus $0.50 per 1000 URL fetches), $7 per 1000 searches for OpenAI, and $14 per 1000 searches for Google.

Perplexity runs before 2026-09-24 recorded search at the standalone Search API rate of $5 per 1000. The cost table corrects those runs by subtracting $2.50 per 1000 recorded search calls, so their totals are lower than in earlier versions of this report. The 2026-09-24 runs use the tool cost that Perplexity reports in each response, which also includes URL fetches.

The prompts are generally short enough that token caching does not occur at all for the Gemini models (needs over 4000 tokens). Some token caching does occur for OpenAI (although these costs do not include that), but it is relatively small (and the majority of token costs are output).

Table 3: Cost and runtime breakdown per full-corpus checker run (36 papers; repeat rows are independent re-runs)

| Model | Provider | Date | Token Cost (USD) | Search Cost (USD) | Total (USD) | Per Paper (USD) | Sec/paper | Wall min |
|---|---|---|---|---|---|---|---|---|
| gemini-3.5-flash-lite | gemini | 2026-07-29 | 6.14 | 8.44 | 14.59 | 0.41 | 128.9 | 15.7 |
| google/gemini-3.1-flash-lite | perplexity | 2026-07-30 | 12.15 | 4.88 | 17.03 | 0.47 | 406.2 | 51.0 |
| google/gemini-3.1-flash-lite | perplexity | 2026-08-03 | 13.43 | 5.86 | 19.29 | 0.54 | 448.7 | 57.9 |
| google/gemini-3.1-flash-lite | perplexity | 2026-08-13 | 7.85 | 2.1 | 9.95 | 0.28 | 339.6 | 41.9 |
| google/gemini-3.1-flash-lite | perplexity | 2026-08-13 | 6.42 | 2.1 | 8.52 | 0.24 | 185.9 | 22.7 |
| google/gemini-3.5-flash-lite | perplexity | 2026-08-18 | 6.67 | 2.01 | 8.68 | 0.24 | 250.9 | 31.6 |
| google/gemini-3.5-flash-lite | perplexity (fast) | 2026-09-24 | 6.76 | 0.94 | 7.69 | 0.21 | 157.6 | 20.2 |
| google/gemini-3.5-flash-lite | perplexity | 2026-09-24 | 7.09 | 2.33 | 9.42 | 0.26 | 173.1 | 22.3 |
| google/gemini-3.7-flash | perplexity | 2026-08-18 | 9.25 | 2.07 | 11.32 | 0.31 | 237.4 | 29.9 |
| google/gemini-3.7-flash | perplexity | 2026-08-21 | 9.85 | 2.1 | 11.95 | 0.33 | 193.7 | 23.8 |
| gpt-5.4-nano | openai | 2026-07-30 | 7.17 | 16.28 | 23.45 | 0.65 | 445.5 | 54.4 |

Table 3: Cost and runtime breakdown per full-corpus checker run (36 papers; repeat rows are independent re-runs)

| Model | Provider | Date | Token Cost (USD) | Search Cost (USD) | Total (USD) | Per Paper (USD) | Sec/paper | Wall min |
|---|---|---|---|---|---|---|---|---|
| gpt-5.6-luna | openai | 2026-07-30 | 6.84 | 11.49 | 18.33 | 0.51 | 328.6 | 39.9 |
| gpt-5.6-luna | openai | 2026-08-03 | 6.9 | 11.53 | 18.43 | 0.51 | 386.2 | 47.1 |
| gpt-5.6-luna | openai | 2026-08-13 | 3.26 | 11.87 | 15.13 | 0.42 | 263.4 | 32.2 |
| gpt-6-luna | openai | 2026-09-22 | 1.62 | 10.87 | 12.49 | 0.35 | 250.9 | 31.0 |
| openai/gpt-5.4-nano | perplexity | 2026-07-30 | 7.34 | 2.57 | 9.91 | 0.28 | 243.3 | 29.7 |
| openai/gpt-5.6-luna | perplexity | 2026-08-01 | 11.21 | 5.32 | 16.53 | 0.46 | 291.6 | 35.7 |
| openai/gpt-5.6-luna | perplexity | 2026-08-13 | 4.04 | 3.02 | 7.06 | 0.2 | 319.4 | 38.9 |
| openai/gpt-6-luna | perplexity (fast) | 2026-09-24 | 1.85 | 1.9 | 3.75 | 0.1 | 349.9 | 44.5 |
| openai/gpt-6-luna | perplexity | 2026-09-24 | 1.58 | 3.84 | 5.42 | 0.15 | 332.4 | 42.4 |
| openai/gpt-6-luna | perplexity (fast) | 2026-11-12 | 1.98 | 1.87 | 3.85 | 0.11 | 384.9 | 48.7 |
| perplexity/nemotron-3.5-lightning-30b-a3b | perplexity | 2026-08-13 | 1.66 | 2.62 | 4.27 | 0.12 | 730.2 | 88.7 |

**Sec/paper** is the mean document wall time from each run's `document_metrics.csv` (how long one paper takes end-to-end under the batch concurrency settings). **Wall min** is full-corpus elapsed time with up to five documents in parallel. Repeat rows again show variance: Perplexity 3.1 flash-lite was about **$17.03 / 51 wall min** and **$19.29 / 58 wall min** in July/early August, then **$9.95 / 42 min** and **$8.52 / 23 min** on 2026-08-13. OpenAI direct `gpt-5.6-luna` was about **$18.3**–**$18.4** with wall time **40**–**47 min**, then **$15.13 / 32 min** on 2026-08-13. The Perplexity 3.7 flash runs were about **$11.32 / 30 wall min** on 2026-08-18 and **$11.95 / 24 wall min** on 2026-08-21 ($0.31 and $0.33 per paper, respectively).

Fast search lowered the search bill in both 2026-09-24 pairs. For Perplexity `openai/gpt-6-luna` it dropped from $3.84 to $1.90 (total $5.42 to $3.75). For `google/gemini-3.5-flash-lite` it dropped from $2.33 to $0.94 (total $9.42 to $7.69). Search-call counts were similar across tiers (1,330 vs 1,365 for luna), so the saving comes from the per-call price rather than fewer searches. GPT-6 Luna is also much cheaper on tokens than earlier models: about $1.6–$1.9 per corpus, compared with $6–$13 for the Gemini and `gpt-5.6-luna` runs. As a result, Perplexity `openai/gpt-6-luna` with fast search is the cheapest configuration tested, at about **$0.10 per paper**, below Nemotron ($0.12). Its wall time (about 44 minutes) is longer than the Gemini flash runs, because luna takes more research steps per citation.

Across this corpus, costs for the earlier configurations were typically 20–50 cents per paper, with web search often a large share of the total. OpenAI direct remains the most search-heavy configuration ($10.87 of the $12.49 `gpt-6-luna` total), which is one reason it is the fallback rather than the primary.

### 5.5 Population Estimates of Precision

The ground truth sample I have collected is likely not representative of the overall population of scholarly papers. I have intentionally selected examples papers that have high rates of hallucinated citations. In Topaz et al. (2026) and Zhao et al. (2026), they estimate that pre-print servers and PubMed papers currently have rates of hallucinated citations ranging from 0.2% to almost 1.9%.

So for an hypothetical example, if false positive rates for hallucinations in my tool are 0.5%, and the population prevalence rate of hallucinations are 1%, and my recall rates are 60%, what are the estimates for precision in the population? The estimated precision in the population will be only slightly over 50%. So out of 100 citations, 1 is a real hallucination, and 99 are not. Overall I capture that real hallucination 60% of the time, so 0.6 true positives. Of the 99 I have `0.99*0.005` false positives, so approximately 0.5 false positives. And then the precision is then `0.6/(0.5 + 0.6)`, which is slightly over 50% (Gigerenzer 2011).

If true rates of hallucinations are lower, my population precision will also be lower. Increasing recall could help, but even if recall is increased to 80% precision is still only 60%.

As such, it will be necessary for humans to carefully review the output of the tool. If an article with a single hallucinated is identified, it does not for sure indicate that AI was used in the production of the bibliography. If multiple references are flagged as hallucinations though, it is more likely there are substantive issues with the work that a dutiful editor should investigate.

Ultimately they overall rate of 1% hallucinations are a mixture of papers – most scientists likely do a good job, but there are likely a few bad apples that use AI to generate whole slews of papers. And those individuals may have hallucination rates in over 20% of their citations (Walters and Wilder 2023). Although with the improvement of the generative AI tools, that will likely decrease over time.

## 6 User Feedback

### 6.1 Peter Moskos: *Back from the Brink*

In August 2026, Peter Moskos reviewed VerusCite's results for his book *Back from the Brink: Inside the NYPD and New York City's Extraordinary 1990s Crime Drop* (Moskos 2025). The original report classified 142 citations as 117 verified, 11 minor errors, 10 not found, and 4

hallucinations. Manual review showed that three of the four hallucination flags were clear false positives.

The remaining hallucination classification was reasonable:

> Krauss, Clifford. 1995. "New York City Crime Falls But Just Why Is a Mystery." *The New York Times*. p. A1. January 1. https://www.nytimes.com/1994/12/26/nyregion/drug-sweep-into-queens-is-postponed.html

The title describes one article, but the supplied URL resolves to a different article. That mismatch is consistent with the hallucination category because the citation combines metadata from two distinct sources.

The checker prompt was updated in response. When searching for historical newspaper and magazine articles, the agent must try multiple search formulations, including the title together with the named publisher (for example, the article title plus "New York Times"), author, and date. It must rely on direct first-party material to verify a citation; indirect sources that merely cite the claimed work are not sufficient. If the direct URL cannot be checked because it is blocked, paywalled, or otherwise inconclusive, the result should be **not found**, without claiming that the citation was fabricated.

PDF link recovery was also improved. In one citation, the visible text layer had collapsed `black-self-sufficiency` into `blackself-sufficiency`, even though the PDF's embedded hyperlink still contained the correct URL. The extractor now reads PDF link annotations and uses the exact embedded target to repair a matching damaged URL in the extracted text. This is conservative: it requires the rest of the URL to match and does not substitute an unrelated link merely because it has the same publisher or hostname.

The new report classifies the same 142 citations as 127 verified, 11 minor errors, 4 not found, and **0 hallucinations**. The interface now presents hallucinations and not found as separate summary categories. It also tells users to manually review hallucination flags because false positives remain possible, and to review not-found results because that category means the system could not make a determination—not that the citation is fabricated.

## 7 Comparison between my tool and other approaches

For some differences between my work and Topaz et al. (2026) and Zhao et al. (2026), my tool can flag as hallucinations when authors or journals are clearly wrong. Both Topaz et al. (2026) and Zhao et al. (2026) focus on title matches (and mismatches) only. The prior example I gave in Canessa et al. (2026) would not be classified as a hallucination in either of these tools, as the title exists – it just clearly swapped out an incorrect journal.

Additionally both Topaz et al. (2026) and Zhao et al. (2026) use google scholar as a secondary reference source. I do not consider this as valid, as I have personally seen google scholar reference hallucinated citations, see this example of a hallucinated reference, the citation

> [5] J. Smith, Spatial analysis of crime patterns using GIS-based decision support tools, J. Quant. Criminol. 38 (2) (2022) 125–148

that is in Kebede et al. (2026) in my ground truth, and actually links to the (false) google scholar citation.

These methods focus on title matches. So for example, Topaz et al. (2026) give the example as a *not* hallucinated title:

> For example, a reference listed as *Depression and anxiety in young adults with ID* corresponds to the real indexed title *Depression and anxiety symptoms during the transition to early adulthood for people with intellectual disabilities* and is probably a reference error, not a fabrication.

My current system, while calling stochastic LLMs, reliably classifies this as a *minor error* (across all model permutations).

Other current tools, like GPTZero or Reviewer3, do not publish how they exactly approach citation checking. While Reviewer3 lists a reference set of metrics, they do not share the actual citations. (It appears these are simulated, as the Walters and Wilder (2023) paper they cite has 636 citations, and Reviewer3's reference set only has 476 citations.)

The articles additionally focus on peer reviewed literature, and do not evaluate hallucinations in non journal articles (grey literature). My benchmark includes the full corpus in the reference papers, including blog posts, working papers, datasets, technical reports, law cases, etc. I have included the not found category for the scenario where a citation cannot be reasonably identified from web based resources, but including these sources will likely increase false positive rates, as how they are even supposed to be correctly cited can be difficult to know exactly.

## 8 Privacy

VerusCite uses only zero data retention (ZDR) models from all providers. Uploaded documents are not used for model training by any third party.

Additionally, the pipeline minimizes what is sent to external LLMs. Citation extraction uses text search on the locally-extracted PDF text first – only the citation strings themselves are sent to an LLM for structured parsing. During verification, only the parsed citation metadata (title, authors, year) is sent to web search, not the full document text. The full PDF content never leaves the server.

## 9 Limitations and Next Steps

The V1 corpus is heterogeneous by design, but 36 documents is a limited sample. The next evaluation (V2) will use the same prompts and pipeline but a **completely different validation set** – different documents, different domains, different sources of known hallucinations. This ensures the system is not overtrained on the particular examples in V1.

Known limitations of the current evaluation:

- Ground truth labels for some documents have ambiguous categories that likely could be reasonably changed. This is particularly true for the minor error category.
- "Not found" is an inherently ambiguous category – some citations are bare URLs, others exist but are difficult to locate via web search (paywalled, very recent, or in non-English databases).
- The corpus over-represents arXiv preprints relative to other domains (law, humanities, clinical medicine).

## 10 Reproducibility

All data for this report is public at github.com/apwheele/veruscite-data. The `v1/` directory contains:

- `ground_truth.csv` – hand-labeled citations with `expected_status`
- `extraction_run/` – raw extraction outputs per model
- `checking_run/` – raw checker outputs per model
- `manifest.json` – run IDs used in this report

To regenerate this report:

```
pip install -r requirements.txt
quarto render report.qmd
```

## 11 AI Use Disclosure

This paper was prepared with AI assistance. Drafting and earlier iterations used **Claude Opus 4.6** (Anthropic), reviewing prior works by Andrew Wheeler (see A. P. Wheeler (2026) for that workflow). Additional edits, including the Current Production Configuration section, multi-run variance notes for independent checker re-runs (Perplexity `google/gemini-3.1-flash-lite` and OpenAI `gpt-5.6-luna`), per-paper runtime in the cost table, the 2026-08-13 re-runs plus Nemotron row, and the 2026-08-18 Perplexity Gemini 3.5 flash-lite / 3.7 flash runs with the production switch to 3.7 flash, were assisted by **Grok 4.5 / 4.6** (xAI). The 2026-08-21 Gemini

3.7 Flash benchmark update and the Peter Moskos user-feedback case study were prepared with **Codex (GPT-5)** (OpenAI). The 2026-09 GPT-6 Luna and Perplexity fast-search benchmark update, including the Perplexity search-price correction, was prepared with **Claude Opus 5.5** (Anthropic). The 2026-11 HALLMARK-driven checker changes, the 2026-11-12 re-run, and the Changelog were prepared with **Claude Fable 5.2** (Anthropic). Ground-truth labels and final review are done by myself (Andrew P. Wheeler). All errors are my own.

# 12 Changelog

## 12.1 2026-11-12: HALLMARK-driven checker changes

I ran VerusCite on the public splits of the HALLMARK benchmark (Kim et al. 2026), 812 valid and 1,260 hallucinated BibTeX entries. The full analysis is at github.com/apwheele/VerusCiteHallmark. Of the valid entries, 767 came back verified. That run showed three errors in the checker, and the production checker changed as follows.

- **Wrong or non-existent venue on a real DOI.** When a DOI resolved and the title and authors matched, the static CrossRef check did not compare the cited venue with the venue on the record. HALLMARK entries that kept a real DOI and swapped in another or an invented venue were verified (41 entries). The DOI path now sends the citation to the web-search checker when the cited venue clearly differs from the record, as the title-search path already did.
- **Wrong year.** A close title and author match was enough to verify, whatever the year. HALLMARK entries dated 2030 to 2036 for papers published in 2021 to 2023 were verified (17 entries). A static match is now a minor error when the cited year is two or more years from every year CrossRef holds for the work (issued, print, online). Any verified citation dated two or more years after the date of the check is also a minor error. A one-year difference is still handled by the online-first versus issue-year rule, and preprints and books are exempt.
- **Author order.** Three valid HALLMARK entries were flagged for author order when the cited order matched the official record. In a repeat test, the checker took the order from a DOI landing page that lists authors alphabetically. The prompt now requires a position-by-position comparison and does not accept re-sorted listings as the byline order, and the checker is told when the cited order matches the CrossRef record.

On the V1 corpus these changes left the 2026-11-12 run within re-run variance of the 2026-09-24 run (see Current Production Configuration). V1 has few citations of these kinds, so the effect shows on HALLMARK and not here: re-checked against CrossRef, all 17 future-date entries became minor errors and 39 of the 41 venue entries are no longer verified by the static check. I have not re-run the full HALLMARK benchmark with the new checker.

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